Adagio Institute, Inc.
Houston, Texas, United States

Doctoral Examination Report

Examination for the Degree of
Doctor of Philosophy in Mathematical Physics
(Concentration in Stochastic Quantum Theory)
by Published Works

Candidate:	Benjamin D. Summers
Doctoral Examiner:	E. F. Boero Adagio Institute, Inc.
Independent evaluation:	Dr S. R. Ismail-Sutton, Durham University
Examination period:	August - September 2026
Report date:	October 2, 2026
Status:	Final examination report (prepared for signature and archival deposit)

This report constitutes the official record of the doctoral examination of the candidate's six-paper portfolio under the Institute's published examination regulations for the degree by Published Works. It states the assessment of the portfolio and the Doctoral Examiner's concluding recommendation. The itemised findings of the technical examinations, the candidate's responses embedded beneath them, and the Doctoral Examiner's dispositions are set out in the accompanying Technical Annex; the review correspondence of record and the other supporting documents are deposited as exhibits.

Contents

1 The examination **2**

1.1 Governance and process 2

1.2 The questions before the examiner and the governing standard 2

1.3 Materials examined 3

1.4 Method and disclosure 3

2 Executive summary and overall recommendation **4**

3 Portfolio assessment **5**

3.1 What the portfolio establishes 5

3.2 Comparison with the ordinary doctoral standard 7

3.3 The identified deficiencies and their resolution 7

3.4 The review exchanges 8

3.5 Contextual information 9

4 Concluding doctoral recommendation **10**

4.1 Answers to the examination questions 10

4.2 The standard applied 10

4.3 Recommendation 11

4.4 Final judgment 11

5 Revision summary **12**

5.1 Corrections made during the examination 12

5.2 Suggested comments 12

6 Certification and signatures **13**

Exhibits **15**

1 The examination

1.1 Governance and process

The examination was conducted under the Institute's published regulations for the degree by Published Works (Adagio Institute, Inc., *Ph.D. in Mathematical Physics with Concentration in Stochastic Quantum Theory: Curriculum, Regulations, and Academic Policies, 2026 Edition.* DOI: 10.5281/zenodo.21370817 (https://zenodo.org/records/21370817)). The process comprised seven steps: (1) publication of the governing regulations; (2) submission of a portfolio of published works satisfying the requirement (here, six papers against a three-preprint minimum); (3) a technical examination of each paper, in two independent reviews: an itemised technical report by the Doctoral Examiner and an independent technical evaluation commissioned by him under Section 5.1 of the regulations from Dr S. R. Ismail-Sutton (Durham University), preceded by a portfolio-level technical evaluation with written questions to the candidate; (4) the candidate's written response to every finding of both reviews, with a revised deposit of each paper; (5) further rounds in which the Doctoral Examiner re-derived the new material, certified each response against the revised text, and made the recommendation recorded in Section 4; (6) resolution by the Board of Academic Directors of Adagio Institute, Inc., from which the candidate is recused and from which the Doctoral Examiner, himself a member of the Board, proposes to recuse himself; and (7) archival of this report, its Technical Annex and its exhibits as part of the permanent academic record, which the Institute may make public (Section 5.3 of the regulations). The Doctoral Examiner determined that the written record provides a sufficient basis for recommendation, and no oral examination was required.

Where the record is. This Report states the assessment and the recommendation. The evidence on which they rest is in the *Technical Annex*, a separate document, and in the *exhibits*. The Technical Annex sets out, for each paper, every finding of both reviews, the candidate's response beneath it and the Doctoral Examiner's disposition, with the record of rounds (Annex, Section T.1), the corrections to the examination record (T.2), the six paper examinations (T.3–T.8) and the full record of revisions (T.9). The *exhibits* are the underlying documents: the review correspondence of record, deposited in full rather than paraphrased; the versions of the papers examined; the technical reports and independent evaluations; the verification record; and the regulations. This Report, the Technical Annex and the exhibits are held in the Institute's GitHub repository, https://github.com/Adagio-Group/Adagio-Institute, which also keeps the record of the examination process; the list at the end of this Report gives the location of each. The Technical Annex and the exhibits form part of the Doctoral Examination Report for the purposes of Section 5.3 of the regulations.

1.2 The questions before the examiner and the governing standard

The examination is directed at two questions, which the concluding recommendation of Section 4 answers explicitly:

Q1. Has the candidate demonstrated original doctoral-level scholarship that makes a coherent contribution to knowledge?

Q2. Do the identified deficiencies materially undermine that demonstration, or are they the kind of issues that are appropriately addressed through revision?

The purpose of doctoral examination is to determine whether the candidate has demonstrated the independent ability to generate original, valuable scholarship at the doctoral level. It is not to determine whether every result survives subsequent criticism unchanged. Accordingly, the examination distinguishes between deficiencies that are evidence against doctoral competence and deficiencies that are the ordinary product of developing original research and are appropriately corrected through revision. The object of evaluation throughout is the candidate's demonstrated research capability; the manuscripts are the evidence. Every finding is classified on that distinction (Technical Annex, Sections T.3–T.8).

The regulations state the same standard in their own terms: the examination determines "whether the submitted portfolio demonstrates doctoral-level originality, independent research ability, and a coherent contribution to knowledge", evaluates "the submitted research program as a whole", and "publication-level revisions identified during examination shall not, by themselves, preclude award of the degree unless they materially undermine the originality, coherence, validity, or doctoral character of the submitted research program" (Published Works, Section 5).

1.3 Materials examined

The portfolio comprises six works, publicly deposited as preprints and examined, at the close of the examination, in the versions of record below (Exhibits B–G). Works deposited by the candidate outside this research programme were not submitted as part of the portfolio and form no part of this examination.

Paper 1. *The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization.* SSRN, DOI 10.2139/ssrn.6922380; earlier version at PhilSci-Archive, ID 28724 under the title *The Nonexistence of the Instantaneous Present* (March 2026). Version of record: 26 September 2026, 14 pp.

Paper 2. *Koopman Spectral Theory for Indivisible Stochastic Processes.* SSRN, DOI 10.2139/ssrn.6733181. Version of record: 25 September 2026, 47 pp.

Paper 3. *Convex Perturbation and the Suppression of the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.6812461. Version of record: 27 September 2026, 18 pp.

Paper 4. *Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation.* SSRN, DOI 10.2139/ssrn.6922758. Version of record: 28 September 2026, 38 pp.

Paper 5. *Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.7119779. Version of record: 18 September 2026, 23 pp.

Paper 6. *Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit.* SSRN, DOI 10.2139/ssrn.7206578, as given in the candidate's letters. Version of record: 24 September 2026, 23 pp.

1.4 Method and disclosure

Each paper was audited line by line, and every computationally checkable claim was re-derived or recomputed independently from the published statements, without use of the candidate's code.

Proofs were checked by hand. Each response of the candidate was certified against the text of the version of record, not against his description of it. A statement is recorded as correct only where it was re-derived or recomputed, and as defective only where the failing sentence and the reason can be exhibited. Errors and omissions of the examination itself, several of them identified by the candidate, are recorded in the Technical Annex, Section T.2; none alters the recommendation.

The Doctoral Examiner independently reviewed the technical examinations, the candidate's responses, and the underlying manuscripts; the terms on which he adopts the findings of this Report are stated in the certification (Section 6).

2 Executive summary and overall recommendation

The six papers develop a mathematical theory of *indivisible* stochastic processes: two-time transition families that violate the Chapman–Kolmogorov (CK) factorization and so admit no time-local generator on the observed state space. The premise that such processes correspond to quantum systems is J. A. Barandes's and is attributed throughout; the candidate supplies the operator-theoretic, perturbative, geometric and statistical apparatus that Barandes does not develop, and the operator-theoretic results stand independently of any reading of indivisibility. Paper 1 argues that completeness of the instantaneous state is a physical assumption which the construction of the state does not supply, and proves a one-sided test of memorylessness. Paper 2 builds a spectral theory of the composition deficit, with a local invariant (the indivisibility tensor), deficit subspaces, a sharp three-pair obstruction to divisible approximation, and a jointly continuous dilation. Paper 3 proves an exact identity for the deficit under convex mixing, which forces a two-threshold theory of detectability. Paper 4 builds a finite-grid estimation theory on path space and determines exactly what it estimates. Paper 5 proves that the natural gauge route to an invariant is empty and builds instead a metric projection theory with a liftability profile and a memory rank. Paper 6 proves an exact balance law for the deficit under coarse-graining.

The examination located defects in every paper, almost all in statements, hypotheses, inferences, citations and one numerical section rather than in load-bearing theorems: in Paper 1, a claim about CK divisibility the arguments did not deliver and a false window claim; in Paper 2, a false witness, an unsatisfiable hypothesis and a misapplied selection theorem, found by the independent evaluator; in Paper 3, a false uniqueness inference and an unsettled model status; in Paper 4, an invalid inference and an inconsistent calibration procedure; in Paper 5, a numerical section that could not establish what it claimed; in Paper 6, statements of scope beyond the theorems. The candidate answered every point of both reviews individually, accepted most, declined some with sound reasons, corrected the examiners where they were wrong, and in several places went beyond what was asked. Every correction is in the versions of record and was certified against the text; each finding, response and disposition is set out in the Technical Annex. The exchanges were substantive on both sides, and the candidate developed several of the reviewers' suggestions into new results (Section 3.4).

Section 4 answers the two examination questions on this record, Q1 in the affirmative and Q2 that the deficiencies do not materially undermine the demonstration and have been addressed through revision, and recommends **Pass**, with waiver of the covering document, a set of suggested comments that are not conditions of the award, and this Report, its Technical Annex and its exhibits archived as the examination record.

3 Portfolio assessment

3.1 What the portfolio establishes

Paper 1. A conceptual paper in the foundations of physics. It argues that dynamical completeness of the instantaneous state -(a uniquely solvable initial-value problem in the deterministic setting, the Markov property in the stochastic one) is a physical assumption that the construction of the state does not supply, and that established physics gives positive reasons to doubt it: the value at a point is the continuous extension of interval descriptions, whose dynamical sufficiency the construction does not guarantee; the Planck-scale and Salecker–Wigner clock bounds obstruct the limit $\delta t \to 0$ within the semiclassical domain where they are derived; and field theory places its content in extended regions (singular coincident-point products, Wightman's theorem for fields of bounded operators, the time-slice axiom). The paper adds one result, Lemma 1: for an irreducible finite-state chain in its stationary law, the window-averaged composition defect vanishes at least linearly in the window width with an explicit constant, while for a non-lumpable observation it converges to the lumped family's defect. This gives a one-sided test of memorylessness on an observed state space. Its originality lies on the foundational axis, where it is moderate; that is the axis on which the candidate asked it to be graded and on which the second round agreed, and he does not claim new physics for it, Lemma 1 being its one piece of new mathematics. It is the premise of the programme. Its Section 7 states the open problem the portfolio leaves. Augmenting the state with its full history renders any process Markov, so a formulation of incompleteness requires a physically fixed class of admissible descriptions, and the paper gives none that is independent of the description. The paper fixes that class only in part, by the resolution bound and the non-invasiveness condition, and neither constraint bounds history depth or state-space dimension. A bound on dimension would exclude the path-space description, but the memory rank of Paper 5, which measures that quantity, is defined relative to a fixed observed state space and transition family, so bounding the class does not by itself make it invariant (Technical Annex, Section T.3).

Paper 2. The operator-theoretic core and the strongest single paper. For two-time families on $L^2(\Omega, \mu)$ without cocycle structure it works through the singular structure of the deficit $\Delta(t, s, r) = P(t, s) - P(t, r)P(r, s)$ rather than through eigenexpansions. Its load-bearing results are the indivisibility tensor $\beta(s) = C(s) - L(s)^2 - L'(s)$ with the exact leading-order law $\Delta(t, s, r) = (t-r)(r-s)\beta(s)+o((t-s)^2)$ (Proposition 4.12); the deficit subspaces and the three-pair obstruction placing every divisible comparison at distance at least $D/3$ under a contraction assumption, now proved sharp (Proposition 4.7); the gauge-relative splitting of the deficit into an interference and a non-composition term, with explicit counterexamples to interpreting either alone (Theorem 4.8); a Stinespring dilation jointly continuous in (t, s) with diagonal retention (Theorem 5.4); and an integrated deficit that vanishes exactly under divisibility (Theorem 7.4). The imported machinery is classical and is now labelled as such; the invariant β and the global deficit locus are the paper's own, and the sharpness of the obstruction was established in exact form in the course of the examination.

Paper 3. A short, correct paper of modest depth. The exact identity $\Delta(\Gamma_g) = (1 - g)[X - gK]$ for convex mixing with a divisible reference makes the non-monotonicity of the response an algebraic fact, which forces a two-threshold formulation of detectability (first crossing and terminal suppression), with bounds attained by an explicit configuration (the attainment was established in the course of the examination). The post-examination Proposition 3.2 (the deficit up to scale δt

is exactly $(1-g)S(g)$ with S convex) is the candidate's own and locates the only possible interior divisible coupling exactly.

Paper 4. The most professionally executed paper and the clearest satisfaction of the reproducibility standard: every reported number was reproduced from the text, and the code and seeds are stated to be deposited. It develops self-normalized importance sampling on finite-grid path space with weights given by a discretized Onsager–Machlup action against a surrogate that is not the truth, proves that the continuous-path form of its own construction is degenerate (Proposition 2.3), and states its estimator theory (standard results, presented as such) in two concentration tiers. After revision it identifies the construction's mechanism exactly: the selected measure is a product of two path densities whose precisions add, so that its quadratic variation differs from the proposal's; from this it derives a necessary design rule for the transfer hypothesis under grid refinement (Proposition 7.5). Its central limitation is stated by the paper itself: the estimand is an expectation under the selected measure, and its relation to a target process holds only under an external hypothesis, attainable under refinement only for designed triples. The identity on which the revised paper rests its novelty claim emerged from the review exchange, and the candidate made it the paper's organizing result.

Paper 5. The most conceptually ambitious paper. Its best idea, and the best single idea of the portfolio, is a negative theorem: the admissible re-dressings drive every family to the flat locus, so no orbital minimization of a flatness distance defines an invariant (Theorem 3.3, Corollary 3.4). The replacement is a metric projection theory (model sets F_d of compressions of d-state flat transports, a liftability profile h_d and a memory rank) with two linear-programming lower bounds, class dependence with an explicit gap, and grid monotonicity. The numerical section, which could not establish a positive profile from upper bounds, was withdrawn and replaced by five exact examples. The stabilization of the model sets at $D_{\max}$ and the joint-law defect were added in the course of the review exchange.

Paper 6. The best-evidenced paper: every displayed quantity is an exact rational or algebraic number, and every one was reproduced. Under a lumping of a family with positive laws it splits the fine deficit orthogonally into a contractive flux and a severed remainder and proves the exact balance law $\Delta_{V^\lambda} = \Phi + P$, with production sourced at the cut; it characterizes exactly when production vanishes (closure on the accessible subspace; for a flat fine family, two-time CK lumpability, which weak lumpability implies but which does not imply it); it proves that the charge is monotone under coarse-graining in neither direction, with exact counterexamples; and it proves tower monotonicity of the deficit norm with both hypotheses shown sharp, one by an explicit perfect-interference construction with a strictly positive joint law (Proposition 6.5). New in revision, and the candidate's own, is a two-sided comparison of the charge with the deficit norm with explicit constants and exponent one (Lemma 2.5, Proposition 2.6), sharpened for law-transporting families in Corollary 2.8 on the examiner's suggestion.

Coherence. Each paper arises from a question the preceding work leaves open. Paper 1 poses the status of CK divisibility; Paper 2 builds the theory of its failure; Paper 3 asks how that failure responds to perturbation, Paper 4 how one estimates on path space without a generator, Paper 5 how one measures distance from divisibility through dilations, and Paper 6 how that measure

transforms under coarse-graining. Papers 3–6 import from Paper 2 the deficit operator, the no-generator proposition, the three-pair obstruction and the integrated deficit, and Paper 6 builds on Paper 5's profile; the examiner verified every imported result. The portfolio is a single research programme, evaluated as such as Section 5 of the regulations requires.

3.2 Comparison with the ordinary doctoral standard

An ordinary Ph.D. is generally meant to establish that the candidate can identify an original research problem; master the relevant literature and mathematics; formulate definitions and conjectures; prove original results; detect and repair errors; communicate the work coherently; and sustain independent research. On this record:

- *Original problem, definitions and results* — evidenced. The programme builds on Barandes's premise what he does not develop, through new objects (the deficit tensor and subspaces, the two thresholds, the liftability profile and memory rank, the production term) and precisely posed open problems.
- *Mastery of the mathematics* — evidenced across operator theory on non-normal families, dilation theory, stochastic analysis and path-space measures, Monte Carlo estimation, real semialgebraic geometry, linear programming, and lumpability and projection-operator theory, including where each tool stops working.
- *Mastery of the literature* — the weakest capacity as submitted: the reviews found priority gaps in five of the six papers. All are now cited and distinguished; further items are suggested in Section 5.2. This is the gap peer review ordinarily closes, and the works had none.
- *Detecting and repairing errors* — mixed, and stated as such. The submitted papers already published negative results against the author's own constructions (Paper 4, Proposition 2.3; Paper 5, Theorem 3.3; Paper 6, Theorem 3.1). During the examination most errors were found by the reviewers; the candidate carried out the repairs, several beyond what was asked, found errors in the examiners' own work, and declined with sound reasons points he judged wrong.
- *Communication and sustained research* — evidenced: claims graded by level of rigour and stated in two registers; every review point adjudicated individually; six papers deposited between March and July 2026 forming one programme, each revised through two or three rounds within four weeks.

The work was produced without a supervisor, outside a university, and deposited without peer review. The examination therefore also served the function that refereeing and supervision ordinarily serve, and the candidate's work under it is part of the evidence (Section 3.4). The standard applied is not lowered.

3.3 The identified deficiencies and their resolution

Table 1 summarizes, paper by paper, the principal deficiencies of the texts first examined and how each was resolved. Every finding, with its severity, the candidate's response and the disposition, is in the Technical Annex, Sections T.3–T.8.

Table 1: Principal deficiencies and their resolution. All resolutions are in the versions of record and were certified against the text.

No.	Principal deficiencies (texts first examined)	Resolution in the version of record	Annex
1	A claim about CK divisibility the arguments do not deliver; a "prediction" that ordinary lumping produces; completeness undefined; a false window claim; an inconsistent clock argument	Original claim restored; Definition 1; prediction replaced by Lemma 1 and a one-sided test; arguments scoped	T.3
2	A false witness (Prop. 2.11(c)); hypothesis (C2) unsatisfiable in infinite dimensions; a misapplied selection theorem; an augmentation claim false as written; priority	Padded witness; (C2) on the open region; corrected completion argument; claim constrained; attributions; sharpness of the bound proved	T.4
3	A false uniqueness inference; the status of the model; an overclaiming abstract; a remark on false hypotheses; an unproved specialization	Exact criterion (Prop. 3.2); mixture as a formal path; claims withdrawn or restated	T.5
4	An invalid inference ("forced"); an inconsistent calibration; a Mori–Zwanzig closure without fluctuating force; literature distinctions that did not hold; a convergence table	Inference withdrawn, with Example 7.4; quasi-likelihood calibration; closure written out; comparison rebuilt; study rerun with code; design rule (Prop. 7.5)	T.6
5	A numerical section offering upper bounds as evidence, not reproducible; an invalid inference; a novelty sentence with prior art	Section withdrawn and replaced by five exact examples; stabilization theorem; joint-law defect; positioning	T.7
6	The metric placed in the companion's class; projection-freeness and charge-level claims beyond the theorems; production called non-negative	Norm equivalence; explicit comparison with exponent one (Prop. 2.6); claims restated	T.8

Classification under the governing standard. None of these deficiencies is evidence against doctoral competence. Each is local (a statement, a hypothesis, an inference, a citation, or one section) and none invalidates a result on which the favourable assessment rests. They are the errors characteristic of unrefereed research in its first year: overreach in abstracts and discussions, hypotheses not checked against the setting, and an incomplete literature. The two most serious, the vacuous Theorem 7.4 of Paper 2 and the numerical section of Paper 5, were repaired without loss to the programme. One matter is a limitation rather than a defect: Paper 4 estimates an expectation under the selected measure, and its link to a target process is a stated open problem. With every defective portion removed or rebuilt, the contributions listed in Section 3.1 stand, each re-derived or recomputed by the examiner.

3.4 The review exchanges

The exchanges between the candidate and the two reviewers were substantive and continued over two or three rounds for every paper. The reviewers proposed constructions, counterexamples and questions; the candidate adjudicated each on its merits, and developed many of them into results

now in the papers; for example the exact sharpness of the three-pair obstruction (Paper 2, Proposition 4.7), the product identity and the design rule of Paper 4, the stabilization theorem and the joint-law defect of Paper 5, the explicit comparison constants of Paper 6, and the statement of the open problem of admissible descriptions that ends Paper 1. This is the ordinary function of rigorous review in the development of research, and the candidate's capacity to take up, test, correct and extend the reviewers' proposals is itself evidence of the independent research ability the examination is concerned with. The central results on which the recommendation of Section 4 rests (the indivisibility tensor and the deficit locus of Paper 2, the deficit identity of Paper 3, the gauge trivialization and the projection theory of Paper 5, and the balance law of Paper 6) were in the papers as submitted. The course of each exchange is set out in the Technical Annex, Sections T.3–T.8.

Declarations of AI assistance. The earlier deposits of Papers 4, 5 and 6 carried a declaration that AI assistance was used in preparing the manuscript and that the frameworks, methods and conclusions are the author's own. The versions of record no longer carry it. Nothing in the examination bears against the candidate's command of the work credited to him: across some twenty letters he answered every technical question in detail, re-derived and extended the examiners' constructions, and found errors in the examiners' own computations. Restoration of a declaration is suggested.

3.5 Contextual information

Section 2 of the published-works regulations requires contextual information for each work. The candidate did not supply it; the Doctoral Examiner has compiled it from the record (Table 2), and the candidate's acknowledgment (Section 6) confirms it.

Table 2: Contextual information for each work, compiled from the record. Every work is sole-authored.

No.	Venue	Citations	Significance
1	SSRN; earlier version PhilSci-Archive	Papers 2–6 only	Premise of the programme; one-sided test of memorylessness
2	SSRN	Papers 3–6 only	Spectral theory of the deficit; the invariant β; sharp obstruction
3	SSRN	Papers 4–6 only	Exact deficit identity under mixing; two thresholds
4	SSRN; code and seeds stated as deposited	Papers 5, 6 only	Path-space estimation without a generator; its mechanism and limits
5	SSRN	Paper 6 only	Gauge trivialization; projection theory and memory rank
6	SSRN	None	Exact balance law for the deficit under coarse-graining

SSRN and PhilSci-Archive are recognized repositories under Section 2, and the portfolio complies. SSRN's readership, however, is principally in finance, economics and law; the communities competent to assess this work read arXiv, and none of the works has appeared in a peer-reviewed venue. The candidate's expertise is demonstrated by the record; his profile within the field, which Section 4 of the regulations asks the examiner to assess, is at present negligible. That is correctable, and a deposit on arXiv is suggested.

4 Concluding doctoral recommendation

4.1 Answers to the examination questions

Q1. Has the candidate demonstrated the independent ability to generate original, valuable doctoral-level scholarship? Yes.

The grounds are drawn from the record, and not from the number of papers. *Original:* the programme extends a premise it attributes to Barandes with an apparatus he does not develop, and the reviews identified no antecedent for its central objects as the papers now state them, all of which were in the papers as submitted: the indivisibility tensor and the global deficit locus (Paper 2), the gauge trivialization (Paper 5), the balance law with orthogonal conservation and its sharpness construction (Paper 6), and the exact deficit identity under mixing (Paper 3), whose two-threshold structure is now stated against the Markovian-admixture measures it refines. Where prior art exists the papers now say so, and the claim that survives is stated at its true size. *Valuable:* these are reusable objects with exact statements, explicit examples and precisely posed open problems, and the negative theorems (Paper 4, Proposition 2.3; Paper 5, Theorem 3.3; Paper 6, Theorem 3.1) close routes a reader would otherwise take. *Independent:* the programme was conceived, executed and deposited without supervision; under examination the candidate adjudicated every point of two reviews on its merits, carried out the repairs himself, developed the reviewers' proposals beyond what they offered (Paper 2, Proposition 4.7; Paper 6, Proposition 2.6 and Corollary 2.8), added results of his own (Paper 3, Proposition 3.2; Paper 2, Section 6.8; Paper 6, Example 4.7), declined with sound reasons what he judged wrong, and found errors in the examiners' own work. *Doctoral-level:* the depth is uneven (Papers 2, 5 and 6 each stand at the doctoral level, Paper 4 is professional work on standard foundations, Papers 1 and 3 are modest) and the portfolio taken as a whole, which is how the regulations require it to be evaluated, meets the ordinary doctoral standard of Section 3.2.

Q2. Do the identified deficiencies materially undermine that demonstration, or are they appropriately addressed through revision? They do not materially undermine it, and they have been addressed through revision.

Localization: every finding classified as major concerned a statement, a hypothesis, an inference, a proof step, a calibration procedure, or one numerical section; none concerned a result on which the answer to Q1 rests. *Survival:* with the defective portions removed or rebuilt, the load-bearing contributions listed in Section 3 stand (Section 3.3), each re-derived or recomputed by the examiner; the two most serious defects (the vacuous Theorem 7.4 of Paper 2; the numerical section of Paper 5) were repaired without loss to the programme. *The evidential weight of the responses:* the candidate's handling of the findings (individual adjudication, correct repairs, reasoned non-adoptions, withdrawals stated as withdrawals) is evidence for, not against, doctoral competence. One matter is not removable by revision and is recorded as such: the link of Paper 4 to a target process is a limitation the paper states and a legitimate open problem, not an error.

4.2 The standard applied

The recommendation below applies the governing standard of Section 1.2: the findings of the paper examinations (Technical Annex, Sections T.3–T.8) were classified by whether they are evidence against doctoral competence or the ordinary product of developing original research, appropriately corrected through revision, and the recommendation rests on that classification. The regulations' own test is met: the publication-level revisions identified do not materially undermine the originality,

coherence, validity or doctoral character of the submitted research programme.

4.3 Recommendation

The Doctoral Examiner recommends that the Institute:

(1) determine that the candidate has satisfied the requirements for candidacy, where procedurally distinct from conferral;

(2) accept the submitted six-paper portfolio, in the versions of record identified in Section 1.3 (Exhibits B–G), as the examined body of work for the degree by Published Works;

(3) waive the covering-document requirement (Regulations, Published Works, Section 5.2), on the grounds that the submitted works themselves provide sufficient integration and context: Papers 2–6 state their place in the programme and cite Paper 1 as its premise, Papers 4 and 5 open with a statement of the programme, every later paper imports its predecessors' results by number, and the synthesis a covering document would supply is given in Section 3 of this report. The ground on which the portfolio-level evaluation recommended against waiver (that Paper 1's misalignment with the later papers needed confronting) has been removed by the revision of Paper 1. Waiver does not reduce the academic standard applied;

(4) record that the corrections identified in the examinations have been incorporated by the candidate into the versions of record, as certified in this report and summarized in Section 5; no correction remains outstanding as a condition;

(5) accept the contextual information compiled in Section 3.5, as confirmed by the candidate, in satisfaction of Published Works, Section 2;

(6) transmit to the candidate the suggested comments of Section 5.2, which are not conditions of the award;

(7) document the examination and revision record, of which this Report, its Technical Annex and its exhibits are the instrument; and

(8) determine the outcome **Pass** and confer the degree.

4.4 Final judgment

The portfolio exceeds the published requirement in number (six works against three) and meets it in substance. It is a single research programme in which each paper arises from a question the preceding work leaves open, carried by results that survive examination, chief among them the indivisibility tensor and the deficit locus of Paper 2, the gauge trivialization and projection theory of Paper 5, and the balance law of Paper 6. The research independence demonstrated is that of a researcher who conceived and executed the programme without supervision and who, under examination, met every objection on its merits, carried out the repairs himself, and corrected his examiners where they were wrong. The deficiencies identified were real, some of them serious, and all of them local; they are corrected in the versions of record. The central results on which this conclusion rests were in the papers as submitted; the exchanges of the examination strengthened them. Nothing remains before conferral beyond the Board's resolution. The suggested comments are offered for the papers' future and for the candidate's standing in his field, not as conditions.

5 Revision summary

5.1 Corrections made during the examination

The candidate made 79 corrections in response to the examinations, all incorporated in the versions of record and certified against their text; none remains outstanding (Table 3). Each is itemised in the Technical Annex, Section T.9, under the closed vocabulary *corrected* (statement or proof repaired), *restated* (claim reduced to what is proved), *withdrawn* (removed, with any successor stated as an open problem) and *qualified* (hypotheses or terminology made explicit).

Table 3: Corrections made during the examination, by paper and disposition.

Paper	**Corr.**	**Restated**	**Withdrawn**	**Qualified**	**Total**	**Outstanding**
1 The Physics of the Instant	5	2	2	3	12	none
2 Koopman Spectral Theory	7	2	0	4	13	none
3 Convex Perturbation	4	2	4	1	11	none
4 Selection Functionals	8	4	3	3	18	none
5 Memory Rank	5	2	2	4	13	none
6 Memory Production	4	2	1	5	12	none
Total	33	14	12	20	79	none

5.2 Suggested comments

The following are offered to the candidate for the future of the papers and of his standing in the field. *They are not conditions of the award*, and the recommendation of Section 4 does not depend on them. Items (S2) and (S3) were not transmitted to the candidate during the examination and are not recorded as unanswered.

(S1) *Declaration of AI assistance.* Restore in Papers 4–6, and consider for Papers 1–3, a declaration of the use of AI assistance, as the earlier deposits carried, or record why it was removed. Journals and repositories increasingly require one.

(S2) *Prior art for Paper 5.* Position the profile against Finesso and Spreij, "Approximate realization of hidden Markov chains" (Proc. IEEE ITW, 2002), which poses the problem of an HMC of given dimension at minimal divergence from a given law; Bacci, Bacci, Larsen and Mardare (ICALP 2017), who prove existence of optimal k-state approximations of a Markov chain and settle the complexity of the threshold problem — does that complexity transfer to h_d?; Qi, Comon and Lim (SIAM J. Matrix Anal. Appl. 37, 2016), who prove closed semialgebraicity of the nonnegative-rank-$\leq r$ set; and Pollock et al. (Phys. Rev. Lett. 120, 040405, 2018), who define non-Markovianity as a minimum distance to Markov process tensors (Paper 2 cites the companion paper of the same authors in Phys. Rev. A; neither is cited in Paper 5).

(S3) *Prior art for Paper 6.* Cite Ledoux, "A geometric invariant in weak lumpability of finite Markov chains" (J. Appl. Probab. 34, 1997), the nearest published analogue of the characterization of Theorem 4.5; acknowledge that the one-measure (π-weighted, χ^2) metric is standard for finite chains, and claim the two-law, two-time extension and the conditional-expectation identification built on it; and engage Teza, Stella and GrandPre (arXiv:2512.11974).

(S4) *Cross-references.* Papers 3–6 cite Paper 2 as "revision 4" (version of record: 6.1); Papers 4–6 cite Paper 3 as "revision 2" (4); Papers 5 and 6 cite Paper 4 as "revision 2" (3.1); Paper 6 cites Paper 5 as "revision 2" (2.1). The result numbers cited are unchanged, but the revision labels should be brought up to date.

(S5) *Deposit venue.* Deposit the six works on arXiv (math-ph, math.PR, quant-ph), where the communities competent to assess them read, and consider submitting Papers 2, 5 and 6 to peer-reviewed journals.

(S6) *Code archive.* Deposit the script of Paper 4 on a persistent archive with its own identifier (for example Zenodo), and cite it from the data statement.

(S7) *Bibliography.* Supply, for the archive, the full bibliography of the candidate's published and deposited works that Section 3 of the regulations attaches to the covering document.

(S8) *Paper 1, editorial.* In Section 4 the sentence after the time-slice sentence ends with "which is the structure this paper argues for". The candidate's letter of 26 September describes that sentence as stating the indexing fact and no more; deleting the clause makes the text match the letter.

6 Certification and signatures

Doctoral Examiner's certification

I have independently reviewed the six versions of record identified in Section 1.3, the texts examined in the first round, the candidate's written responses as set out in the Technical Annex, the independent technical evaluations of the six papers commissioned under Section 5.1 of the regulations from Dr S. R. Ismail-Sutton (Durham University), the verification record and the correspondence of record, and I have certified each of the candidate's responses against the text of the version of record. The independent technical evaluations are incorporated in this report through the findings adopted in the Technical Annex, where her principal points are recorded with the candidate's responses and my dispositions; her evaluations are deposited in full as Exhibit J. By my signature I adopt the findings of this report, the answers to the two examination questions, and the recommendation of Section 4, and I submit the report to the Board of Academic Directors of Adagio Institute, Inc. for resolution.

Ezequiel Boero — October 9th, 2026

E. F. Boero — Date
Doctoral Examiner, Adagio Institute, Inc.

Candidate's acknowledgment

I acknowledge the findings of this Report and its Technical Annex; I confirm that the responses embedded in the Technical Annex are mine, that the versions of record identified in Section 1.3 are the official submitted versions of the papers, and that the contextual information of Section 3.5

is accurate; I have received the suggested comments of Section 5.2; and I recuse myself from the board's resolution of this examination.

Benjamin D. Summers Candidate	Date

Exhibits

The Technical Annex and the exhibits are held in the Institute's GitHub repository, https://github.com/Adagio-Group/Adagio-Institute, at the locations given below, and form part of the Doctoral Examination Report for the purposes of Section 5.3 of the regulations. The repository also keeps the record of the examination process.

Technical Annex.
Examination record prepared with this Report: rounds of the examination (T.1); corrections to the examination record (T.2); the examinations of Papers 1–6, with findings, responses and dispositions (T.3–T.8); record of revisions (T.9).
Repository: 06_final_examiner_report

Exhibit A.
Review correspondence of record for Papers 1–6: electronic mail among the candidate, the Doctoral Examiner and the independent evaluator, 2–29 September 2026, filed by paper and date (Technical Annex, Section T.1).
Repository: 09_procedural_correspondence

Exhibit B.
Paper 1: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P01

Exhibit C.
Paper 2: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P02

Exhibit D.
Paper 3: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P03

Exhibit E.
Paper 4: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P04

Exhibit F.
Paper 5: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P05

Exhibit G.
Paper 6: original version submitted and subsequent revision.
Repository: 01_candidate_submission/papers_all_versions/P06

Exhibit H.
Portfolio-level technical evaluation.
Repository: 02_technical_evaluations

Exhibit I. The Doctoral Examiner's technical reports on Papers 1–6.
Repository: 02_technical_evaluations

Exhibit J.
Independent technical evaluations of Papers 1–6 by Dr S. R. Ismail-Sutton.
Repository: 02_technical_evaluations

Exhibit K.

The governing regulations.
Repository: 00_governance